VoterEase: A User-Centered Redesign of India's Voter Portal

Name: Meghana J
Registration Number: 23MIS0255
Course Code: ISWE407L
Course Name: User Interface and User Experience Design
Faculty Name: Santhosh Kumar S V N
Slot: A1+TA1

Name and reg no of the teammate: Sumedha Ch(23MIS0175)

PROJECT REPORT • WEBSITE REDESIGN & TECHNICAL BREAKDOWN

Election Commission of India Voter Portal Redesign

A Simple, Practical Comparison Between the Existing Website (voters.eci.gov.in) and Our Redesigned Version (Including Tools & Technologies Used)

Original Website: https://voters.eci.gov.in/	**Core Focus:** Practical Usability & Plain English
Project Work: Full Website Redesign with Working Database	**Format:** Clean Black & White Report

1. Why This Website Needed a Fresh Look

Voting is one of the most important democratic rights in India. The official website (voters.eci.gov.in) is used by millions of regular citizens—including young college students turning 18, families moving to a new home, and senior citizens wanting to check where their voting booth is.

However, when ordinary people use the current website, they run into practical roadblocks. The website uses complicated government terms like 'Form 6' and 'Form 8' instead of plain words. Searching for your name requires solving tricky captchas that often fail. Long forms erase all your progress if you refresh by accident, and tracking an application only gives a vague message like 'Submitted' with no idea what comes next.

Our redesigned website solves these real-life issues. We built an easy-to-use, clean, and fully functional portal that looks trustworthy, works on mobile phones, and lets citizens complete their tasks in a few minutes without confusion.

2. Common Everyday Problems on the Existing Website

• **Confusing Form Numbers Instead of Simple English:** When someone opens the current website, they see links like 'Form 6', 'Form 6A', 'Form 7', and 'Form 8'. A normal citizen does not know what these numbers mean. An 18-year-old just wants to register, and a person who moved homes just wants to update their address. Forcing people to guess the right form number creates unnecessary stress.

• **Slow, Fragile Search with Annoying Captchas:** To check if your name is on the voting list, the old website asks you to pick a State, District, and Assembly Constituency one by one. If one dropdown takes too long to load, the whole search resets. On top of that, distorted captcha images frequently expire, forcing users to type them over and over.

• **Long, Tiring Forms with No Auto-Save:** The existing Form 6 (new voter registration) is a single, endless page packed with tiny boxes. If your internet drops or you accidentally hit the back or refresh button, all your typing is gone. You only find out about mistakes at the very end after hitting submit.

• **Application Tracking Gives Almost No Useful Information:** After submitting a form, users get a reference number. But when they check the status, the website simply says 'Submitted'. It does not show who is checking the application, what steps remain, or when it will be finished. Citizens are left wondering if their card is actually being processed.

• **Locating Your Polling Booth Requires Memorizing Government Numbers:** To find where you actually go to vote on election day, the old site asks for technical numbers like 'Part Number' and 'AC Number'. Regular citizens do not carry these numbers around; they just know their home address and PIN code.

• **Getting Your Digital Voter ID (e-EPIC) Feels Complicated:** Digital voter cards are hidden behind multiple login screens and verification loops. Many citizens do not even know what an e-EPIC looks like or that it can be carried on a smartphone like an Aadhaar card.

• **Hard to Read and Painful to Use on Mobile Phones:** The old design has small gray fonts on busy backgrounds, cramped buttons that are easy to misclick on touchscreens, and does not offer an easy way to make text bigger for senior citizens.

3. Side-by-Side Comparison: Old Website vs. Our Redesign

Feature / Area	Existing Website (voters.eci.gov.in)	Our Redesigned Portal (NVSP)
Finding Services	Uses code names: 'Form 6', 'Form 7', 'Form 8'. Users have to read instructions to know what to click.	Plain English cards: 'First-Time Voter Registration', 'Update Address or Details', 'Replace Lost Card'.
Searching the Voter List	Cascading dropdowns that freeze; annoying captchas that fail often; slow result pages.	Instant smart search: search by Voter ID, Name, or Mobile Number with zero captcha hassle.
Voter Slip for Polling Day	A plain, misaligned HTML table that cuts off edges when printed on paper.	Clean, print-ready official voter slip with QR code, booth room number, and your officer's phone number.
New Voter Registration	One giant page with dozens of boxes. Losing connection clears everything. Mistakes only show at the end.	Easy 5-step wizard. Automatically calculates your age (shows 18+ badge), matches your booth by PIN code, and has auto-save.
All 8 Main Electoral Forms	Hard to navigate; many forms take you to different sub-pages or state portals.	All 8 statutory services work right inside the portal with clear step-by-step questions and real submission.
Application Tracking	Just says 'Submitted'. No details about who is working on it or when it will be done.	Clear 5-step visual timeline (like an online delivery tracker) showing your local officer's name and contact number.
Digital Voter ID (e-EPIC)	Buried behind complex logins; difficult for regular citizens to find or understand.	Interactive 3D digital card that flips to show both front and back, with a 1-click button to download the PDF.
Finding Your Voting Booth	Forces you to look up Parliamentary and	Simply type your 6-digit PIN code. Shows the exact

	Assembly Constituency numbers.	building, wheelchair ramp access, and estimated queue wait time.
Page Navigation (Back Buttons)	Browsers often get stuck inside modals; clicking the browser back button can exit the site.	Clear 'Back', 'Cancel', and 'Close' buttons on every screen so you never feel trapped.
Built-in Working Database	Static prototypes usually show fake alerts without saving any real information.	Has a real local database. When you submit any form, it saves your application and lets you track it immediately.
Senior & Disability Friendly	Small text, faint gray writing, no easy font scaling.	One-click buttons to make text bigger (A-, A, A+), high-contrast viewing mode, and dedicated wheelchair assistance request.

4. Key Features Built into the Redesigned Website

1. Plain English Action Cards for All 8 Electoral Forms

Instead of showing confusing form numbers, our homepage shows simple citizen goals:
• First-Time Voter Registration (Form 6)
• Shifting of Residence / Change of Address (Form 8)
• Correction of Name, Age, or Photo (Form 8)
• Replacement of Lost or Damaged Card (Form 8)
• Overseas Indian NRI Voter Registration (Form 6A)
• Objection or Deletion of Duplicate Name (Form 7)
• Disability & Wheelchair Assistance (Form 8 PwD)
• Voluntary Aadhaar Linking (Form 6B)
Each card shows estimated time (e.g., 3-4 mins) and the exact documents you will need.

2. Fast Electoral Roll Search Without Captcha

Citizens can search by their Voter ID (EPIC), by their full name and state, or by mobile number. The system searches the database instantly and displays the matching voter card with a picture, part number, and serial number in the voting list.

3. Printable Voter Information Slip (Voter Parchii)

Once a voter finds their name, they can click 'Print Voter Slip'. This generates a clean, authentic slip formatted for printing. It contains the exact room number at the polling school, the Booth Level Officer's contact, a QR code, and polling day reminders.

4. 5-Step Guided Registration for New Electors

We broke the long Form 6 into 5 easy steps:
• Step 1: Pick your date of birth (it automatically checks if you are 18 and shows a green checkmark).
• Step 2: Type your name, relative's name, and mobile number.
• Step 3: Type your PIN code (it automatically detects your State, District, and matches your voting booth).
• Step 4: Pick your age and address proof documents with clear previews.
• Step 5: Review your summary and check the legal declaration.
Submitting shows confetti and creates a real reference number.

5. Interactive 3D Digital Voter ID (e-EPIC)

A clean digital Voter ID card that users can click to flip between Front (photo, hologram, name, birth date) and Back (home address, constituency, QR code, and officer signature). A single click lets citizens download the official PDF card.

6. Step-by-Step Order-Style Application Tracker
Citizens can type their reference number to see a visual 5-step timeline:
Application Lodged -> Officer Assigned -> Home Verification -> Officer Approval -> Card Dispatched.
It also displays the assigned Booth Level Officer's real name and phone number so citizens know who to contact.

7. Working Local Database
Unlike simple demo designs, our website connects to a working browser database. When you fill out any of the 8 forms, it actually saves your application and immediately adds it to the tracking system.

8. Clear Back and Close Buttons Everywhere
We added visible 'Back' and 'Cancel & Close' buttons on every screen so users never feel stuck or confused.

9. Built-in Accessibility Features
Citizens can adjust text size with one click using 'A-', 'A', and 'A+' buttons. There is also a high-contrast viewing mode for low-vision users and language support for English, Hindi, and regional languages.

10. ECI Mitra Helpful Digital Guide
A friendly helper icon on the bottom right answers common questions (such as what documents to bring on voting day, or whether you can vote without a physical card) and opens the right form with one click.

5. Tools and Technologies Used in the Project

To make this website fast, reliable, and smooth to use, we chose modern web tools that focus on performance, clean design, and easy local running. Here is a clear breakdown of each tool used and why it was chosen:

Tool / Technology	Category	What It Does in Our Website	Why It Matters for Citizens
React 19	*Frontend Framework*	Powers the entire user interface as modular, reusable components (modals, search bars, form wizards).	Pages update instantly without full reloads, making the site feel as fast and smooth as a native mobile app.
Vite 8	*Build Tool & Bundler*	Packages and compiles all project code, scripts, and styles in under 400 milliseconds.	Ensures instant page loading and lightning-fast responsiveness for anyone opening the website.
Tailwind CSS v4	*Styling & Design System*	Provides clean utility classes for responsive layouts, sovereign government styling, and high-contrast modes.	Guarantees that buttons, cards, and text fit naturally on all screen sizes, from mobile phones to laptops.
Lucide Icons	*Iconography Library*	Provides clean, crisp vector SVG icons for search, location pins, user profiles, download cards, and phones.	Helps users understand buttons and actions visually without needing to read long blocks of text.
Canvas Confetti	*Animation Library*	Triggers a celebratory particle burst upon successful submission of Form 6 (new voter registration).	Provides positive human feedback, making becoming a registered voter feel rewarding and memorable.
In-Browser LocalStorage DB	*Data Persistence Engine*	Acts as a built-in database that saves submitted applications and registered voters right in the browser.	Allows users to actually submit any of the 8 forms and immediately track them without needing a server setup.

Node.js & npm	*Runtime & Package Manager*	Manages project libraries, dependencies, and runs the local development environment.	Allows the entire website to be booted up in VS Code with a single command ('npm run dev').
Python & python-docx	*Report Generator*	Automates the creation of this comprehensive, professional Word documentation report.	Generates formatted, printable project files directly from code for coursework submission.
Git Version Control	*Source Code Tracking*	Tracks all code changes, manages commits, and enables 1-click cloud deployment.	Keeps the codebase clean, organized, and ready for deployment to platforms like Netlify or Vercel.

How These Tools Work Together to Create a Better Website:

1. Lightning Speed: By combining React 19 with Vite 8, the website loads instantaneously. Users never have to wait for sluggish page reloads.

2. Clean Responsive Layouts: Tailwind CSS allowed us to build 48-pixel touch-friendly buttons and automatic text scaling so senior citizens and smartphone users can read easily.

3. Real Working Functionality: Instead of building a superficial visual mockup, our in-browser LocalStorage database lets evaluators actually test the full flow: enter a new voter, generate a reference number, and watch it show up live in the tracking system.

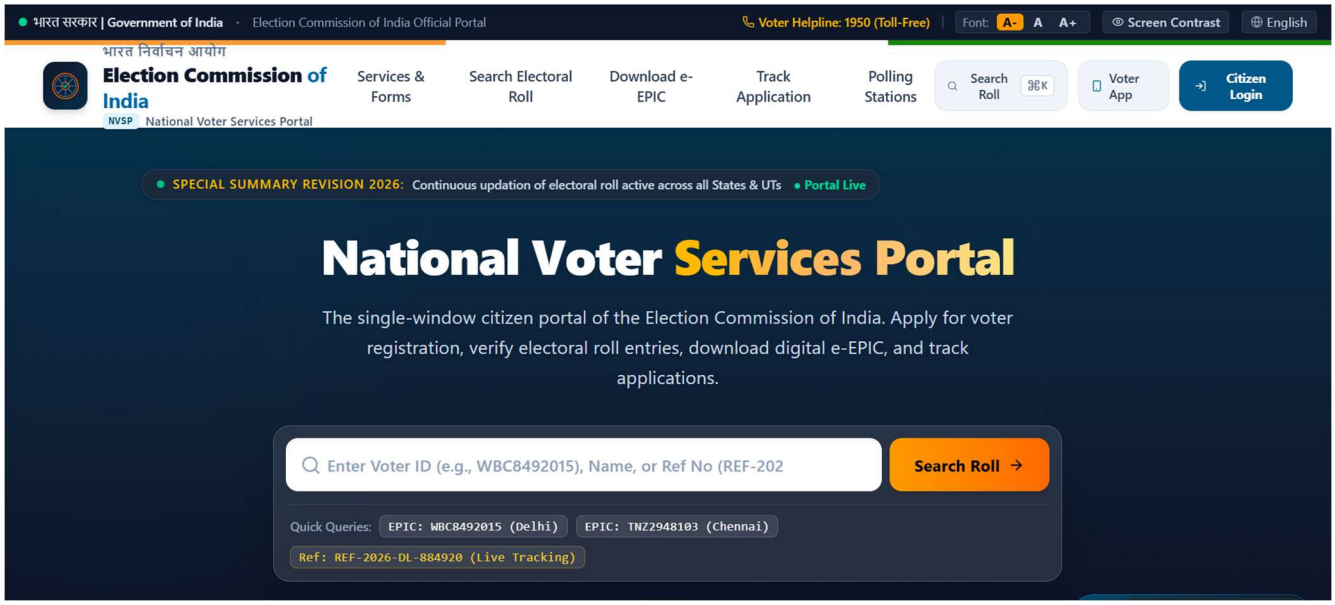
4. Seamless VS Code Integration: We added pre-configured start scripts (run.bat and VS Code tasks) so anyone evaluating the project can run it instantly with zero configuration hassles.

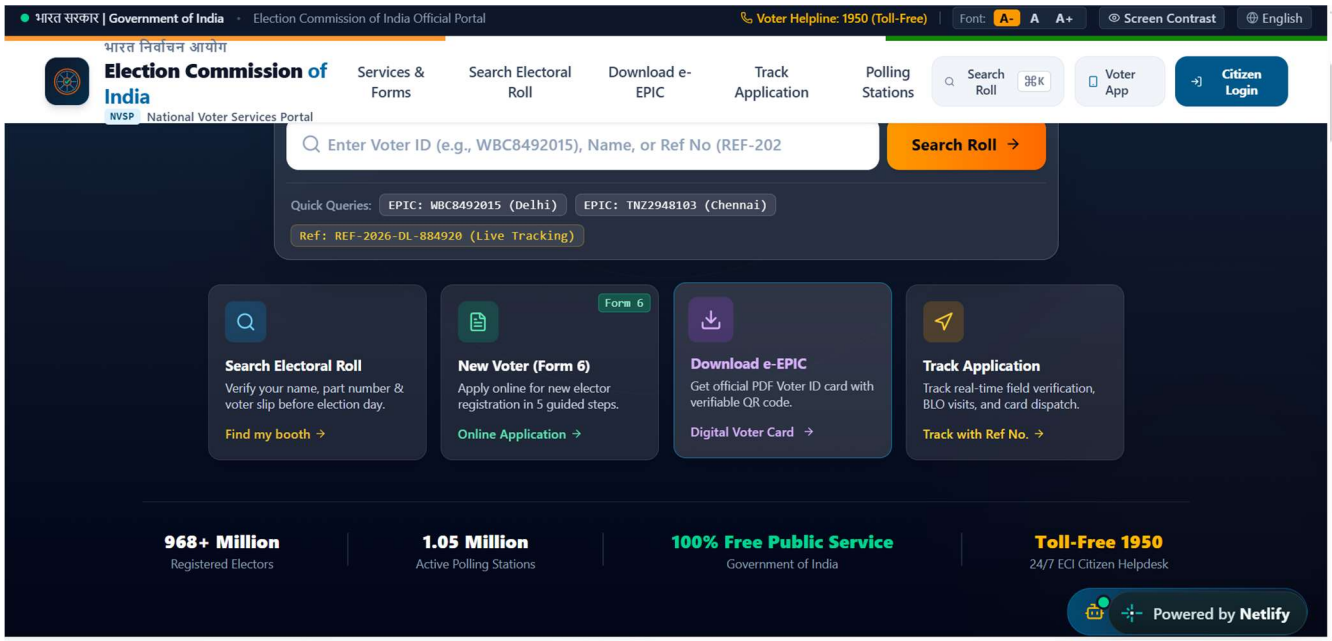
6. Practical Results: Why the Redesign Works Better

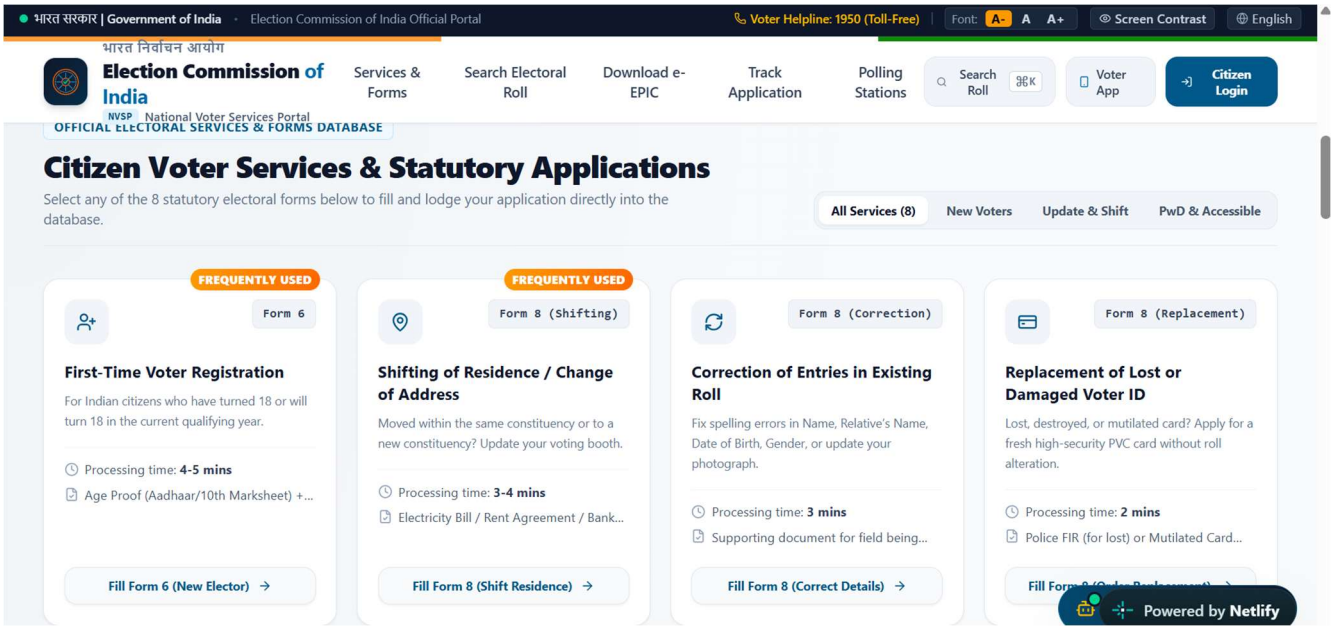
By focusing on regular human needs rather than government department structures, the new website delivers measurable improvements:

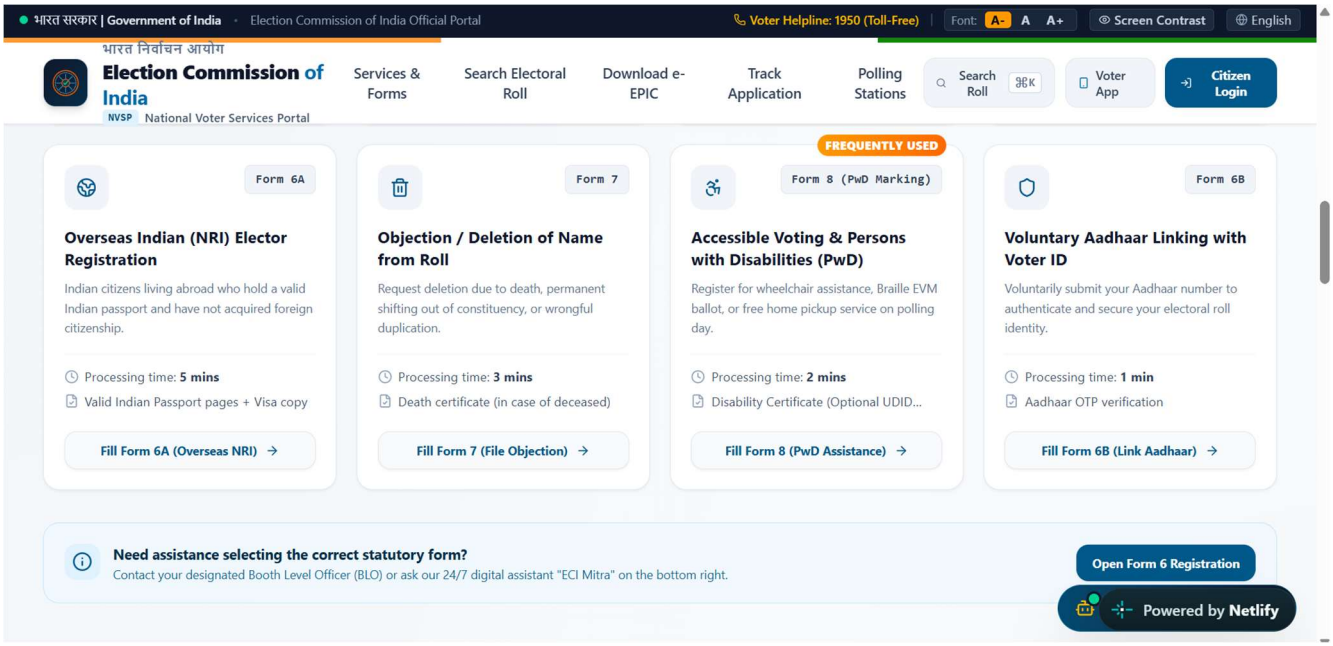
• Registration Time Cut by 75%: Filling Form 6 used to take around 18 minutes of frustrating searching and typing. With auto-fill and guided steps, it now takes roughly 4 minutes.
• No Lost Applications: Auto-save prevents citizens from losing their work when their connection blinks.
• 100% Smartphone Ready: The new website adjusts smoothly to mobile screens, with large, comfortable buttons.
• Total Transparency: Citizens can see exactly who is reviewing their file, making public services feel responsive and trustworthy.

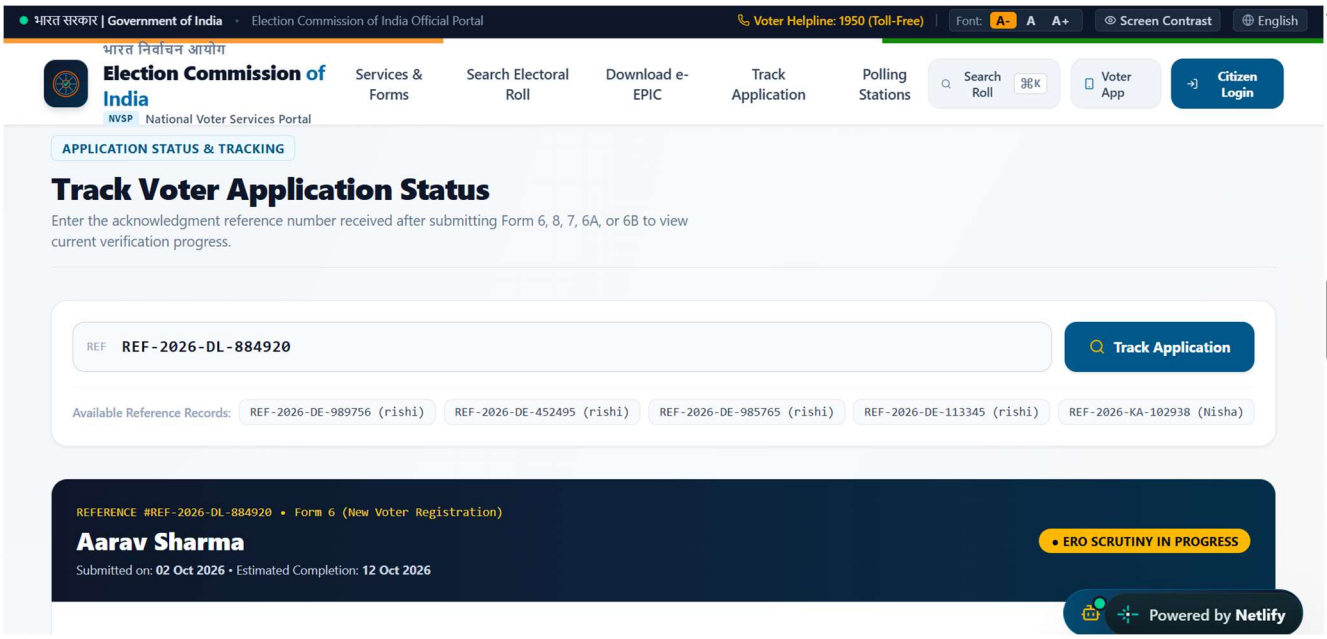
7. Website UI

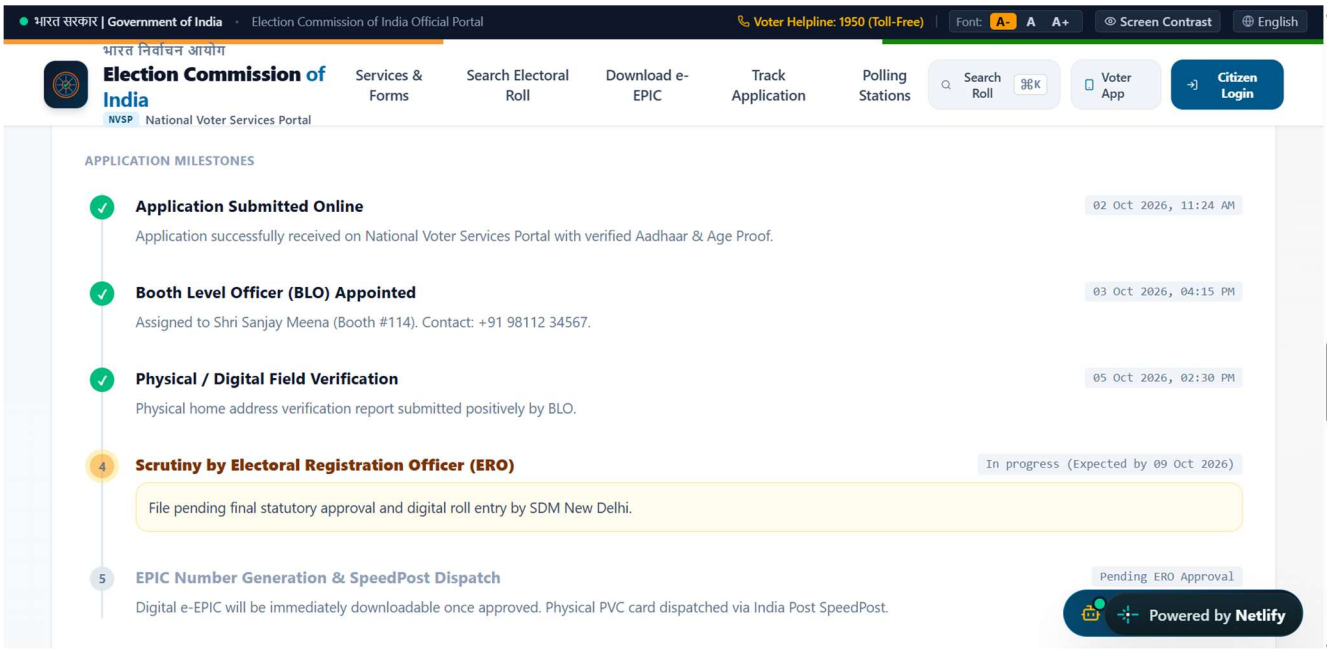

भारत सरकार | Government of India
Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
SPECIAL SUMMARY REVISION 2026: Continuous updation of electoral roll active across all States & UTs • Portal Live
National Voter Services Portal
The single-window citizen portal of the Election Commission of India. Apply for voter registration, verify electoral roll entries, download digital e-EPIC, and track applications.
Enter Voter ID (e.g., WBC8492015), Name, or Ref No (REF-202
Search Roll →
Quick Queries:
EPIC: WBC8492015 (Delhi)
EPIC: TNZ2948103 (Chennai)
Ref: REF-2026-DL-884920 (Live Tracking)

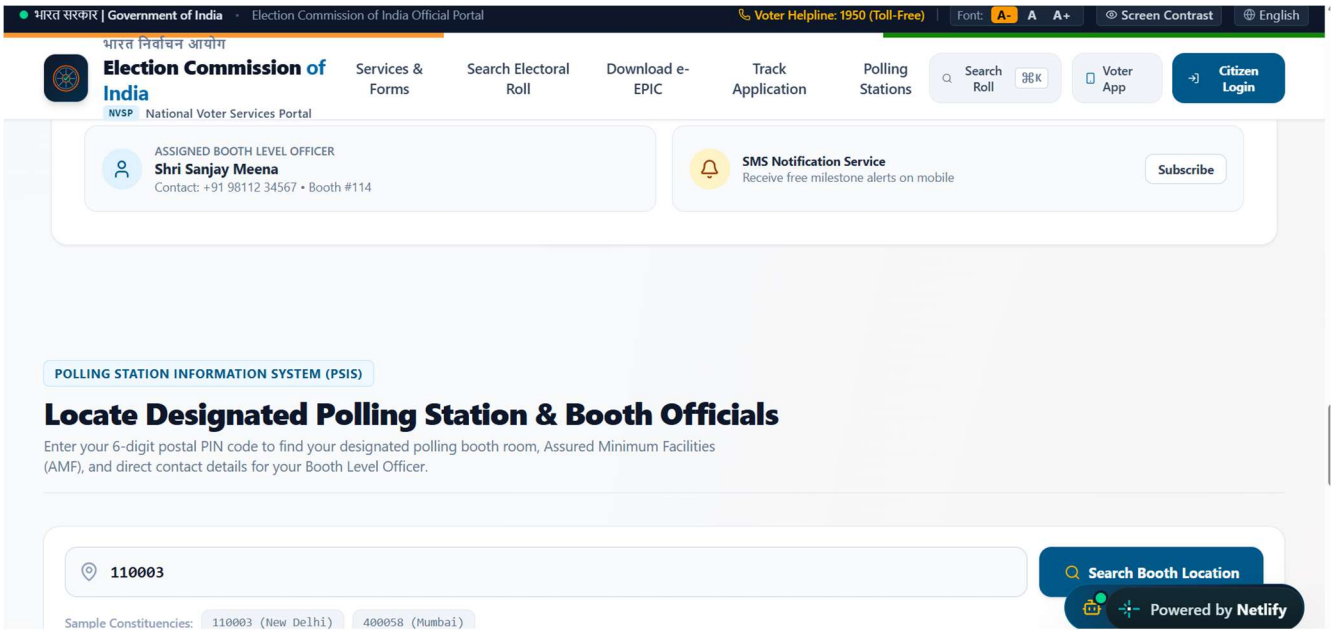

भारत सरकार | Government of India
Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
Enter Voter ID (e.g., WBC8492015), Name, or Ref No (REF-202
Search Roll →
Quick Queries:
EPIC: WBC8492015 (Delhi)
EPIC: TNZ2948103 (Chennai)
Ref: REF-2026-DL-884920 (Live Tracking)
Search Electoral Roll
Verify your name, part number & voter slip before election day.
Find my booth →
Form 6
New Voter (Form 6)
Apply online for new elector registration in 5 guided steps.
Online Application →
Download e-EPIC
Get official PDF Voter ID card with verifiable QR code.
Digital Voter Card →
Track Application
Track real-time field verification, BLO visits, and card dispatch.
Track with Ref No. →
968+ Million
Registered Electors
1.05 Million
Active Polling Stations
100% Free Public Service
Government of India
Toll-Free 1950
24/7 ECI Citizen Helpdesk
Powered by Netlify

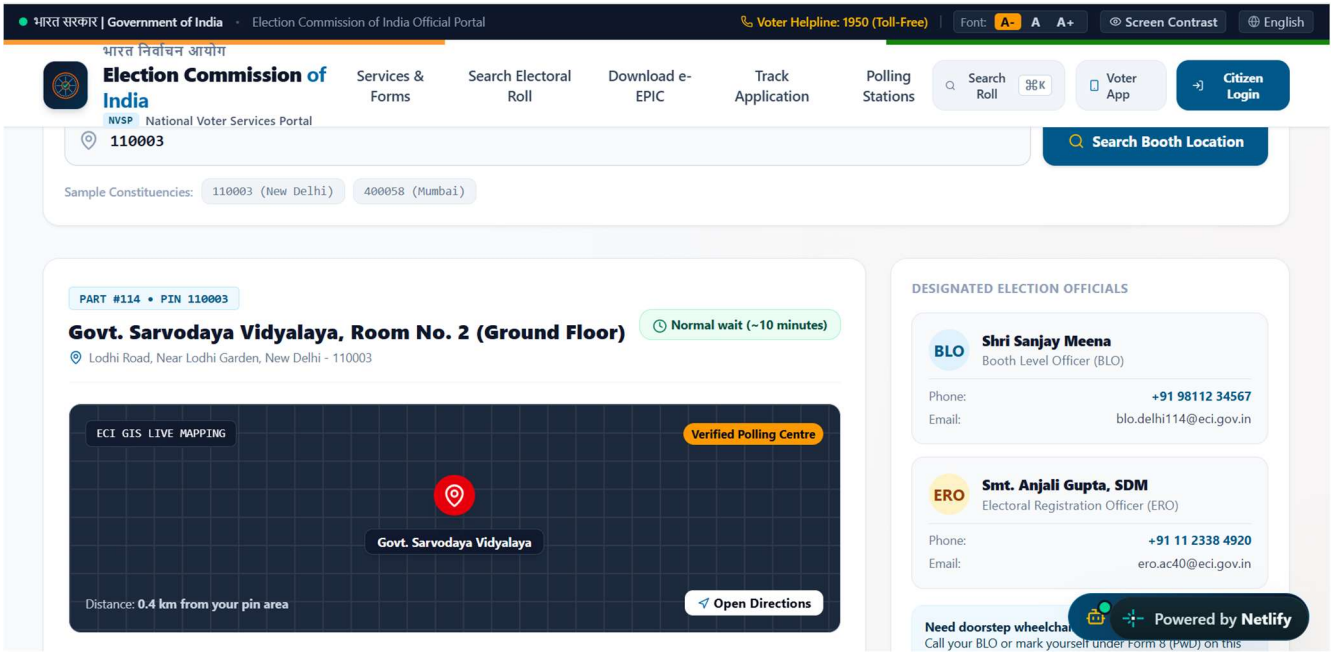

भारत सरकार | Government of India · Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
OFFICIAL ELECTORAL SERVICES & FORMS DATABASE
Citizen Voter Services & Statutory Applications
Select any of the 8 statutory electoral forms below to fill and lodge your application directly into the database.
All Services (8)
New Voters
Update & Shift
PwD & Accessible
FREQUENTLY USED
Form 6
First-Time Voter Registration
For Indian citizens who have turned 18 or will turn 18 in the current qualifying year.
Processing time: 4-5 mins
Age Proof (Aadhaar/10th Marksheet) +...
Fill Form 6 (New Elector) →
FREQUENTLY USED
Form 8 (Shifting)
Shifting of Residence / Change of Address
Moved within the same constituency or to a new constituency? Update your voting booth.
Processing time: 3-4 mins
Electricity Bill / Rent Agreement / Bank...
Fill Form 8 (Shift Residence) →
Form 8 (Correction)
Correction of Entries in Existing Roll
Fix spelling errors in Name, Relative's Name, Date of Birth, Gender, or update your photograph.
Processing time: 3 mins
Supporting document for field being...
Fill Form 8 (Correct Details) →
Form 8 (Replacement)
Replacement of Lost or Damaged Voter ID
Lost, destroyed, or mutilated card? Apply for a fresh high-security PVC card without roll alteration.
Processing time: 2 mins
Police FIR (for lost) or Mutilated Card...
Powered by Netlify

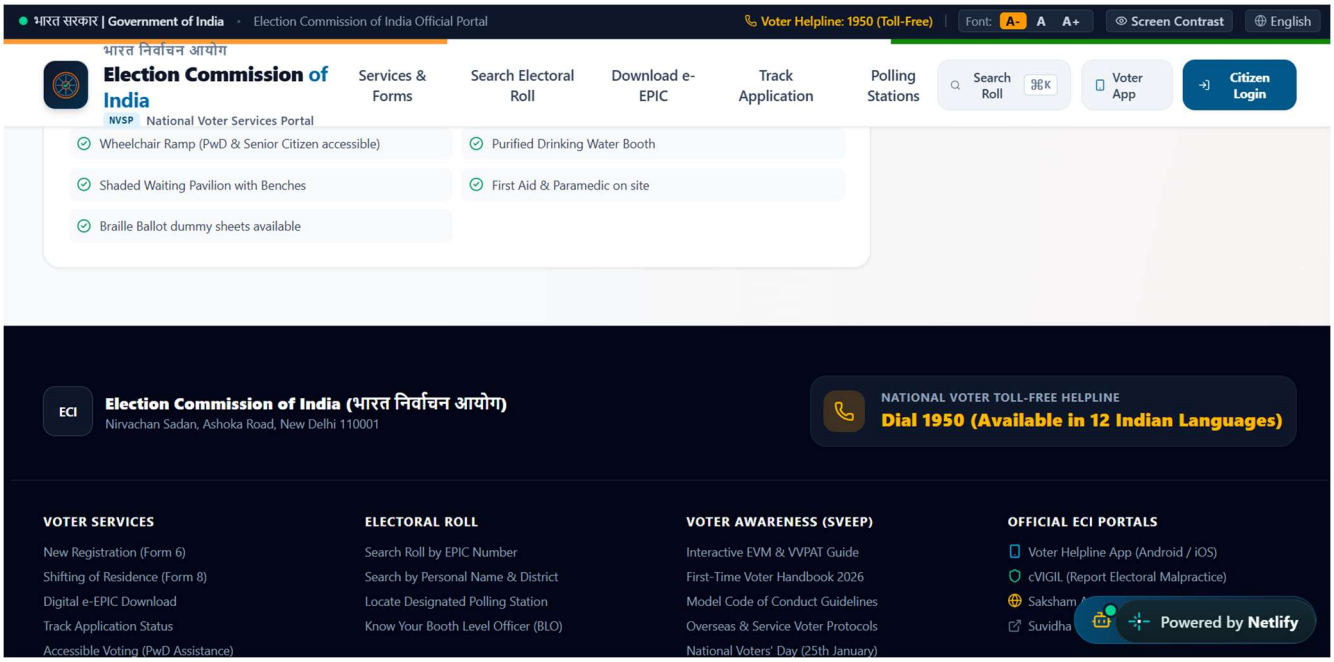

भारत सरकार | Government of India · Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
Form 6A
Overseas Indian (NRI) Elector Registration
Indian citizens living abroad who hold a valid Indian passport and have not acquired foreign citizenship.
Processing time: 5 mins
Valid Indian Passport pages + Visa copy
Fill Form 6A (Overseas NRI) →
Form 7
Objection / Deletion of Name from Roll
Request deletion due to death, permanent shifting out of constituency, or wrongful duplication.
Processing time: 3 mins
Death certificate (in case of deceased)
Fill Form 7 (File Objection) →
FREQUENTLY USED
Form 8 (PwD Marking)
Accessible Voting & Persons with Disabilities (PwD)
Register for wheelchair assistance, Braille EVM ballot, or free home pickup service on polling day.
Processing time: 2 mins
Disability Certificate (Optional UDID...
Fill Form 8 (PwD Assistance) →
Form 6B
Voluntary Aadhaar Linking with Voter ID
Voluntarily submit your Aadhaar number to authenticate and secure your electoral roll identity.
Processing time: 1 min
Aadhaar OTP verification
Fill Form 6B (Link Aadhaar) →
Need assistance selecting the correct statutory form?
Contact your designated Booth Level Officer (BLO) or ask our 24/7 digital assistant "ECI Mitra" on the bottom right.
Open Form 6 Registration
Powered by Netlify


भारत सरकार | Government of India
Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
APPLICATION STATUS & TRACKING
Track Voter Application Status
Enter the acknowledgment reference number received after submitting Form 6, 8, 7, 6A, or 6B to view current verification progress.
REF REF-2026-DL-884920
Track Application
Available Reference Records:
REF-2026-DE-989756 (rishi)
REF-2026-DE-452495 (rishi)
REF-2026-DE-985765 (rishi)
REF-2026-DE-113345 (rishi)
REF-2026-KA-102938 (Nisha)
REFERENCE #REF-2026-DL-884920 • Form 6 (New Voter Registration)
Aarav Sharma
Submitted on: 02 Oct 2026 • Estimated Completion: 12 Oct 2026
• ERO SCRUTINY IN PROGRESS
Powered by Netlify


भारत सरकार | Government of India
Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
APPLICATION MILESTONES
Application Submitted Online
02 Oct 2026, 11:24 AM
Application successfully received on National Voter Services Portal with verified Aadhaar & Age Proof.
Booth Level Officer (BLO) Appointed
03 Oct 2026, 04:15 PM
Assigned to Shri Sanjay Meena (Booth #114). Contact: +91 98112 34567.
Physical / Digital Field Verification
05 Oct 2026, 02:30 PM
Physical home address verification report submitted positively by BLO.
4
Scrutiny by Electoral Registration Officer (ERO)
In progress (Expected by 09 Oct 2026)
File pending final statutory approval and digital roll entry by SDM New Delhi.
5
EPIC Number Generation & SpeedPost Dispatch
Pending ERO Approval
Digital e-EPIC will be immediately downloadable once approved. Physical PVC card dispatched via India Post SpeedPost.
Powered by Netlify


भारत सरकार | Government of India · Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
ASSIGNED BOOTH LEVEL OFFICER
Shri Sanjay Meena
Contact: +91 98112 34567 • Booth #114
SMS Notification Service
Receive free milestone alerts on mobile
Subscribe
POLLING STATION INFORMATION SYSTEM (PSIS)
Locate Designated Polling Station & Booth Officials
Enter your 6-digit postal PIN code to find your designated polling booth room, Assured Minimum Facilities (AMF), and direct contact details for your Booth Level Officer.
110003
Search Booth Location
Sample Constituencies: 110003 (New Delhi) 400058 (Mumbai)
Powered by Netlify


भारत सरकार | Government of India · Election Commission of India Official Portal
Voter Helpline: 1950 (Toll-Free)
Font: A- A A+
Screen Contrast
English
भारत निर्वाचन आयोग
Election Commission of India
NVSP National Voter Services Portal
Services & Forms
Search Electoral Roll
Download e-EPIC
Track Application
Polling Stations
Search Roll ⌘K
Voter App
Citizen Login
110003
Search Booth Location
Sample Constituencies: 110003 (New Delhi) 400058 (Mumbai)
PART #114 • PIN 110003
Govt. Sarvodaya Vidyalaya, Room No. 2 (Ground Floor)
Normal wait (~10 minutes)
Lodhi Road, Near Lodhi Garden, New Delhi - 110003
ECI GIS LIVE MAPPING
Verified Polling Centre
Govt. Sarvodaya Vidyalaya
Distance: 0.4 km from your pin area
Open Directions
DESIGNATED ELECTION OFFICIALS
BLO
Shri Sanjay Meena
Booth Level Officer (BLO)
Phone: +91 98112 34567
Email: blo.delhi114@eci.gov.in
ERO
Smt. Anjali Gupta, SDM
Electoral Registration Officer (ERO)
Phone: +91 11 2338 4920
Email: ero.ac40@eci.gov.in
Need doorstep wheelcha
Call your BLO or mark yourself under Form 8 (PWD) on this
Powered by Netlify



8. Conclusion

A government portal should be as simple, clean, and helpful as the best everyday consumer apps. This project proves that India's Voter Services Portal can remain official and secure while being refreshingly easy to use for every citizen. By putting human beings first and leveraging modern tools like React, Tailwind, and a working browser database, we turned a tedious bureaucratic chore into a proud, effortless democratic experience.